Grama Jyothi Rural Electricity Co-operative
Task 2

TASK 2 – Compare Expectations with Execution (K4)

File to save as: Task02/types_measure.c

```
#include <stdio.h>
#include <limits.h>
#include <float.h>
#include <stdint.h>
#include <inttypes.h>
int main(void)
{
    printf("sizeof(int) = %zu, range = %d to %d\n",
```

sizeof(int), INT_MIN, INT_MAX);

```
    printf("sizeof(long) = %zu, range = %ld to %ld\n",
```

sizeof(long), LONG_MIN, LONG_MAX);

```
    printf("sizeof(long long) = %zu, range = %lld to %lld\n",
           sizeof(long long), LLONG_MIN, LLONG_MAX);
    printf("sizeof(float) = %zu, FLT_DIG = %d\n", sizeof(float), FLT_DIG);
    printf("sizeof(double) = %zu, DBL_DIG = %d\n", sizeof(double), DBL_DIG);
    printf("sizeof(int64_t) = %zu\n", sizeof(int64_t));
    /* Demonstration: add the sample bill in integer paise and float rupees. */
    const int bill_count[] = {4000, 60000, 800000};
    const int64_t sample_bill_paise = 48000; /* Rs 480.00 per household */
    for (size_t i = 0; i < sizeof bill_count / sizeof bill_count[0]; ++i) {
        int64_t total_paise = (int64_t)bill_count[i] * sample_bill_paise;
        float total_rupees = (float)bill_count[i] * 480.00f;
        printf("%d bills: paise total = %" PRId64
```

", float total = %.2f rupees\n", bill_count[i], total_paise, total_rupees);

```
    }
    return 0;
}
```

Compile and Run

```
gcc -std=c11 -Wall -Wextra -pedantic Task02/types_measure.c -o Task02/types_measure
./Task02/types_measure
```

This program uses Rs 480.00 as a controlled example bill for every household, so the scale comparison is repeatable. It does not claim every household will actually have that bill. Record the exact output from your system. Analysis to Complete After Running Paise total: integer multiplication gives an exact total as long as the chosen integer type does not overflow. Float total: float has limited significant decimal precision. Compare the printed value with total_paise / 100.0. int, long and long long: use the printed ranges to decide whether a worst-case district total in paise fits. Do not actually overflow a signed type; signed overflow is undefined behaviour in C. sizeof: record the byte sizes reported by your compiler. C guarantees minimum ranges, not identical byte sizes on every platform. double alternative: double usually has much wider range and better precision than float, but it is still binary floating point and is not a substitute for exact integer paise when exact currency is required. For this fixed-bill example, the exact totals are calculated mathematically in the table; the float observation must come from your run.